

	External Assessment Specifications Document			
Curriculum Code	Qualification Title		NQF Level	
251201002	Occupational Certificate: Artificial Intelligence Software Developer		5	
	Name	Email	Phone	Logo
Assessment Quality Partner	MICT SETA	Gugu.Sema@mict.org.za	011-2072600	

DQP Representative Signature

Date

AQP Representative Signature

Date

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1 ASSESSMENT STRATEGY

1.1 Assessment Model

An external integrated summative assessment, conducted through the relevant QCTO Assessment Quality partner is required for the issuing of this qualification. The external integrated summative assessment will focus on the exit level outcomes and associated assessment criteria.

The assessment will be conducted through written assessment and the evaluation of practical tasks at decentralised approved assessment sites by a panel of assessors evaluated by assessor(s) registered with the AQP within a period of 1 day.

1.2 Qualification Purpose

The purpose of this qualification is to prepare a learner to operate as an Artificial Intelligence Software Developer

Artificial Intelligence Software Developers build AI functionality into software applications through integrating and implementing AI algorithms and logic into the deliverables of an IT project. Developers teach the machine to solve problems the way a human would through the use of programming. They create, test, and deploy code. These developers also assist in converting machine learning APIs so that other applications can use them

A qualified learner will be able to:

- Interpret solution design documentation and develop AI solution
- Train the AI model through a machine learning process and test the performance to ensure that model accuracy is strictly maintained within the selection framework
- Deploy the AI solution and maintain the solution to ensure model accuracy is strictly maintained

1.3 Assessment Standards

1.3.1 Assessment Standards for the Qualification

Integrated Assessment Focus Area 1

Gather and interpret data from various sources to define an AI solution to a real-life world problem (5%)

Associated Assessment Criteria

- Sources from where data can be collected are identified and the quantitative and qualitative quality of such data is verified.
- A critical thinking- and problem solving process is implemented through which the data can be reviewed for the purpose of arriving at an informed conclusion and its signification and implications are identified.
- The scale of measurement for the data is decided on as this will have a long-term impact on data interpretation ROI.
- Data is prepared by cleaning, moving, checking, and organising such data using the appropriate tools.

Integrated Assessment Focus Area 2

Critically analyse data and create a solution design document (SDD) that defines an artificial intelligent (AI) solution that solve a real-life word problem (15%)

Associated Assessment Criteria

- A query is written to extract data from an operational platform and placed in a flat file (coma separated file/CSV file) or in a spreadsheet to determine what the data looks like, what type of data it is and how it ties up with other tables.
- A software tool is used to look at and interpret data structure and the relationship between data.
- Datasets (resultant data from a query) are analysed using the appropriate tools and appropriate criteria:
 - Is the data accessible
 - Will new datasets be added over time
 - How much effort is required to clean/organise data into a usable set
- A Software Design Document (SDD) is created that will describe the AI solution envisioned for the real-life world problem identified through data analysis.

Integrated Assessment Focus Area 3

Choose a type or category of AI learning and the relevant algorithm to analyse data, gain insight and make subsequent prediction, or create a determination with it (10%)

Associated Assessment Criteria

- Taking the data analysed into consideration, the problem type is determined and classified to match the data set (groupings of data either at source or destination) to the identified problem.
- The type of problem being solved, such as classification, regression, anomaly detection or dimensionality reduction is determined and the AI algorithm that work best for each type of problem is determined.
- One of four basic machine learning types is applied to determine the algorithm that should be used, i.e. supervised-, unsupervised-, semi-supervised- and reinforced learning.
- The level of visibility needed in the AI solution is decided on and either decision trees or neural networks are chosen.
- A few different algorithms are used and compared to determine which delivers the most accurate results and an algorithm is selected for employment.

Integrated Assessment Focus Area 4

Train the AI model through a machine learning process and ensuring that model accuracy is strictly maintained within the selection framework. Set a minimum acceptable threshold and apply statistical discipline. Validate the model. (25%)

Associated Assessment Criteria

- The chosen algorithm is trained through machine learning by using the data and by incrementally improving the predictions within the selection framework.
- Accuracy within the selected framework is maintained by setting minimal acceptable thresholds for the application and applying statistical discipline in the training of the algorithm to ensure accuracy.
- The AI system is trained and retrained to ensure that the algorithm achieve the desired accuracy.
- The trained models are tested through cross-validation or by splitting the dataset (70:30) of which a portion of the data is devoted to training and the rest to testing.

- The model is validated using the chosen metric or combination of metrics to measure the objective performance of a model.

Integrated Assessment Focus Area 5

Select a machine learning system and build an artificial intelligent (AI) solution to a real-life world problem. (25%)

Associated Assessment Criteria

- Python's framework and libraries are utilised to solve common programming tasks and simplify the development process.
- The machine learning libraries to be used in building the AI model are determined, taking into account that a good set of libraries means less time writing the algorithm and more time actually building the AI model.
- A machine learning platform that will ease the machine learning process and facilitate building the models is selected to build an AI system.

Integrated Assessment Focus Area 6

Implement and run the AI solution on a selected platform and check the prediction results in real-life use. Select and run the AI solution on a platform. (20%)

Associated Assessment Criteria

- The implementation process planned, prepared and executed before implementation by creating a plan to describe all tasks to be completed.
- The ongoing risk of implementing a new solution is minimised by keeping, maintaining and monitoring a risk schedule.
- Functional resources to key locations are deployed to provide on-site support at an implementation.
- The AI solution is launched ensuring the implementation is operating according to set outcomes and criteria.
- Support is provided at implementation and knowledge on whether intended solution is achieved is shared.

1.3.2 Assessment Standards for Phases

None

1.3.3 Assessment Standards for Part Qualifications

None

2 ASSIGNMENTS TO BE EVALUATED EXTERNALLY

None

3 CRITICAL ASPECTS OF THE INTERNAL ASSESSMENTS TO BE ASSESSED EXTERNALLY

- None

4 CRITERIA FOR THE REGISTRATION OF ASSESSORS

Qualifications:

- Assessors and moderators: Accredited by the MICT SETA
- Subject matter experts: NQF 6 ICT related qualification

Experience:

- 1 year's experience in the IT industry
- 1 year's experience in assessment and moderation

5 FOUNDATIONAL LEARNING

Not required.

6 ELIGIBILITY REQUIREMENTS FOR THE EXTERNAL ASSESSMENT

In order to qualify for the external summative assessment learners must provide proof of completion of all required modules by means of a statement of results and statement of work experience as well as proof of successful completion of the following subjects and modules or alternative programs where applicable, including Foundational Learning Competence.

6.1 Qualification

	<i>Proof of Knowledge Modules</i>			<i>OR</i>			<i>Proof of Alternative Programmes</i>	
<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>		<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>
251201-002-00-KM-01	Overview of Artificial Intelligence	4	2					
251201-002-00-KM-02	Introduction to Mathematics and Statistics	4	10					
251201-002-00-KM-03	Analytical Thinking and Problem Solving	4	3					
251201-002-00-KM-04	Data, Databases and Data Visualisation	4	8					
251201-002-00-KM-05	Computing Theory	4	8					
251201-002-00-KM-06	Introduction to Artificial Intelligence, Machine Learning, Deep Learning	4	5					
251201-002-00-KM-07	Artificial Intelligence	5	12					
251201-002-00-KM-08	Machine Learning	5	16					

251201-002-00-KM-09	Deep Learning	5	16					
251201-002-00-KM-10	Introduction to Governance, Legislation and Ethics	4	1					
251201-002-00-KM-11	Fundamentals of Design Thinking and Innovation	4	1					
251201-002-00-KM-12	4IR and Future Skills	4	4					

	<i>Proof of Practical Skill Modules</i>			<i>OR</i>			<i>Proof of Alternative Programmes</i>	
<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>		<i>Number</i>	<i>Title</i>	<i>NQF Level</i>	<i>Credits</i>
251201-002-00-PM-01	Mathematics and Statistics for Programming	4	8					
251201-002-00-PM-02	Problem Definition, Analytical Thinking and Decision-Making	4	2					
251201-002-00-PM-03	Access, Analyse and Visualise Structured Data Using Spreadsheets	4	4					
251201-002-00-PM-04	Use SQL to Communicate with a Database	5	4					

251201-002-00-PM-05	Build a simple AI solution using Python	5	8					
251201-002-00-PM-06	Use Python Data Scraping to Populate Database Table in SQL	5	4					
251201-002-00-PM-07	Use Machine Learning to Build an AI solution in Python	5	6					
251201-002-00-PM-08	Use Deep Learning to Build an AI Neural Network Architecture in Python	5	10					
251201-002-00-PM-09	Use Deep Learning to Build an AI Neural Network Architecture in TensorFlow	5	10					
251201-002-00-PM-10	Function Ethically and Effectively as a Member of a Multidisciplinary Team	4	3					
251201-002-00-PM-11	Participate in a Design Thinking for Innovation Workshop	4	4					

And

Statement of Work experience

6.2 Phase 1

No Phase Tests applicable

7 PART QUALIFICATIONS

None